

SMART Survey Processing Pipeline and Machine Learning-Based Mortality Prediction

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SMART Survey Processing Pipeline and Machine Learning-Based Mortality Prediction

1. Introduction

Standardized Monitoring and Assessment of Relief and Transitions (SMART) surveys are widely used in humanitarian and public health settings to collect reliable information about population health conditions, including nutritional status, household characteristics, and mortality indicators. SMART methodology provides a standardized approach for conducting surveys in emergency contexts, allowing organizations to generate comparable health information and support evidence-based decision-making (SMART Initiative, 2017).

Mortality estimation is an important component of SMART surveys because it helps measure the severity of humanitarian crises and identify vulnerable populations. Mortality indicators support governments, health organizations, and humanitarian agencies in prioritizing resources and designing effective interventions, especially in areas affected by conflict, food insecurity, and disease outbreaks (World Health Organization [WHO], 2021).

However, SMART survey data requires extensive processing before it can be used for advanced analysis. Raw survey files may contain different structures, formats, missing values, and inconsistent variables. Therefore, data extraction, cleaning, transformation, and integration are necessary to create reliable datasets suitable for analysis and prediction (Kotsiantis et al., 2006).

Traditional mortality analysis methods provide valuable information but may not fully capture complex relationships between multiple health and household variables. Machine learning techniques have become increasingly useful in healthcare research because they can identify patterns from complex datasets and support prediction tasks (Rajkomar et al., 2019).

Machine learning applications in healthcare include risk assessment, disease prediction, and health outcome forecasting. However, mortality prediction remains challenging because mortality datasets are usually imbalanced, with fewer mortality cases compared with non-mortality cases. Therefore, appropriate preprocessing and evaluation methods are required to develop reliable prediction models (Chawla et al., 2002).

This study develops a SMART survey processing pipeline and applies machine learning algorithms for mortality prediction. The research processes raw SMART survey files, extracts mortality and anthropometric information, prepares a structured dataset, and evaluates three classification models: K-Nearest Neighbors (KNN), Support Vector Machine (SVM), and Decision Tree. The aim is to determine whether SMART survey information can be used to predict mortality outcomes and support humanitarian health decision-making.

2. Problem Statement

Mortality prediction from SMART survey data is an important challenge in humanitarian and public health environments because timely identification of mortality risks can support better decision-making and faster responses. In emergency situations, mortality information is needed to understand the severity of a crisis, identify vulnerable populations, and guide the allocation of healthcare resources. However, collecting, processing, and analyzing mortality information from SMART surveys remains a challenging task because survey datasets are complex and require multiple preparation stages before they can be used effectively (SMART Initiative, 2017).

SMART surveys provide valuable information about households, population characteristics, nutritional status, and mortality events. However, the raw survey files are commonly stored in specialized formats and contain different structures depending on the survey type and design. These differences make manual processing difficult and time-consuming. Without proper data processing, important relationships between household characteristics, anthropometric measurements, and mortality outcomes may not be identified.

Another major problem is the difficulty of predicting mortality outcomes using traditional analysis approaches. Mortality events are relatively rare compared with non-mortality cases, creating a highly imbalanced dataset. In such situations, machine learning models may achieve high accuracy by predicting the majority class while failing to identify actual mortality cases. Therefore, developing models that can effectively detect mortality cases requires appropriate preprocessing methods, class balancing techniques, and suitable evaluation metrics (Chawla et al., 2002).

In addition, humanitarian organizations often require rapid access to reliable information to support emergency interventions. Traditional mortality analysis methods usually depend on completed surveys and manual statistical analysis, which may delay the identification of high-risk populations. A machine learning-based approach can provide additional support by automatically analyzing large amounts of survey data and identifying patterns associated with mortality risk.

The main problem addressed in this study is the lack of an automated pipeline that converts raw SMART survey files into a structured dataset suitable for mortality prediction. This involves several challenges, including survey classification, data extraction, missing data handling, variable transformation, dataset integration, and prediction model development.

Therefore, this research aims to solve these challenges by developing a SMART survey processing pipeline and evaluating machine learning models for mortality prediction. The study investigates whether information collected from SMART surveys, including household characteristics and anthropometric measurements, can be used to predict mortality outcomes. The results can contribute to improving the use of SMART survey data and supporting faster evidence-based humanitarian decision-making.

3. Methodology

This study followed a structured methodology consisting of SMART survey data processing, dataset preparation, and machine learning model development. The methodology was designed to transform raw SMART survey files into a clean dataset suitable for mortality prediction. The overall process included survey organization, data conversion, cleaning, integration, feature preparation, model training, and evaluation.

3.1 SMART Survey Data Processing

The first stage involved processing the raw SMART survey files provided in .as format. Each survey file was examined to identify important information, including survey location, survey date, survey level, and survey type. The surveys were classified into different categories based on their structure, including individual and aggregate surveys.

The files were organized into appropriate folders to improve accessibility and ensure that similar survey types were processed together. This step was necessary because SMART surveys may contain different structures depending on the survey design and collection method (SMART Initiative, 2017).

3.2 Data Conversion and Mortality Extraction

After organizing the survey files, the mortality sections were extracted from the SMART survey files and converted into CSV format. Individual and aggregate mortality tables were processed separately because they contain different data structures.

The individual mortality files contained household-level information with multiple household members, including demographic characteristics and mortality indicators. Aggregate mortality files contained summarized mortality records. Separate extraction methods were applied to correctly process both formats and maintain data consistency.

3.3 Data Cleaning and Preparation

The extracted datasets were cleaned before machine learning analysis. The cleaning process included removing duplicate records, converting date fields into standard formats, converting numerical variables into appropriate data types, and checking missing values.

Missing values were handled through preprocessing techniques to ensure that machine learning algorithms could process the dataset correctly. Categorical variables were also transformed into numerical representations because most machine learning models require numerical input features (Kotsiantis et al., 2006).

3.4 Dataset Integration

After cleaning the individual datasets, anthropometric and mortality information were merged using common household identifiers. The merging variables included:

- Survey source file
- Cluster number
- Team number
- Household number

A final machine learning dataset was created containing household characteristics and the mortality prediction target variable.

The target variable was created as a binary classification problem:

- **0 = No recorded mortality**
- **1 = Mortality recorded**

This allowed the machine learning models to learn patterns associated with mortality outcomes.

3.5 Feature Preparation

The final dataset contained demographic and anthropometric features used for prediction. The selected variables included:

- Sex
- Age in months
- Weight
- Height
- MUAC
- WAZ
- HAZ
- WHZ
- Household information

Categorical features were encoded into numerical values. Feature scaling was applied using standardization to ensure that variables with different measurement ranges contributed appropriately during model training.

Because mortality cases represented a small proportion of the dataset, the Synthetic Minority Oversampling Technique (SMOTE) was applied to balance the training data. SMOTE creates synthetic samples of the minority class and improves model learning in imbalanced classification problems (Chawla et al., 2002).

3.6 Machine Learning Models and Evaluation Process

After preparing the final dataset, machine learning models were developed to predict mortality outcomes. This study selected three classification algorithms from the available machine learning methods: K-Nearest Neighbors (KNN), Support Vector Machine (SVM), and Decision Tree. These algorithms were selected because they provide different learning approaches and are commonly used for binary classification problems.

K-Nearest Neighbors (KNN)

K-Nearest Neighbors is a supervised learning algorithm that classifies new observations based on the similarity between the observation and existing data points. The model identifies the nearest examples in the training dataset and assigns the class based on the majority class among those neighbors (Cover & Hart, 1967).

In this study, KNN was selected because mortality prediction involves identifying households with characteristics similar to previously observed mortality cases.

Support Vector Machine (SVM)

Support Vector Machine is a supervised learning algorithm that separates different classes by finding an optimal decision boundary between them. SVM attempts to maximize the margin between classes, which can improve classification performance, especially in complex datasets (Cortes & Vapnik, 1995).

SVM was selected because it is effective for binary classification tasks and can handle complex relationships between variables.

Decision Tree

Decision Tree is a supervised learning algorithm that creates a structure of decision rules by repeatedly splitting data according to feature values. The model produces an interpretable tree structure that shows how different variables contribute to classification decisions (Breiman et al., 1984).

Decision Tree was selected because it provides easier interpretation compared with many other machine learning algorithms and allows analysis of feature importance.

Model Training

The dataset was divided into training and testing groups using an 80/20 train-test split. The training dataset was used to develop the models, while the testing dataset was used to evaluate their performance on unseen data.

The training data was balanced using SMOTE to reduce the effect of class imbalance. Each model was trained using the balanced training dataset, and predictions were generated using the testing dataset.

Evaluation Metrics

The performance of the three models was evaluated using multiple metrics. Accuracy, Recall, F1-score, Area Under the Receiver Operating Characteristic Curve (AUC), and Confusion Matrix were used.

Accuracy measures the overall percentage of correct predictions made by the model.

Recall measures the ability of the model to correctly identify actual mortality cases. In mortality prediction, recall is especially important because failing to identify high-risk cases can have serious consequences.

F1-score combines precision and recall into a single measurement and provides a balanced evaluation when dealing with imbalanced datasets.

AUC measures the model's ability to distinguish between mortality and non-mortality classes across different classification thresholds.

Confusion Matrix provides detailed information about correct and incorrect predictions by showing true positives, true negatives, false positives, and false negatives.

Using multiple evaluation metrics provides a more complete understanding of model performance because accuracy alone may be misleading when one class is much smaller than the other (Chawla et al., 2002).

4. Results

4.1 Introduction

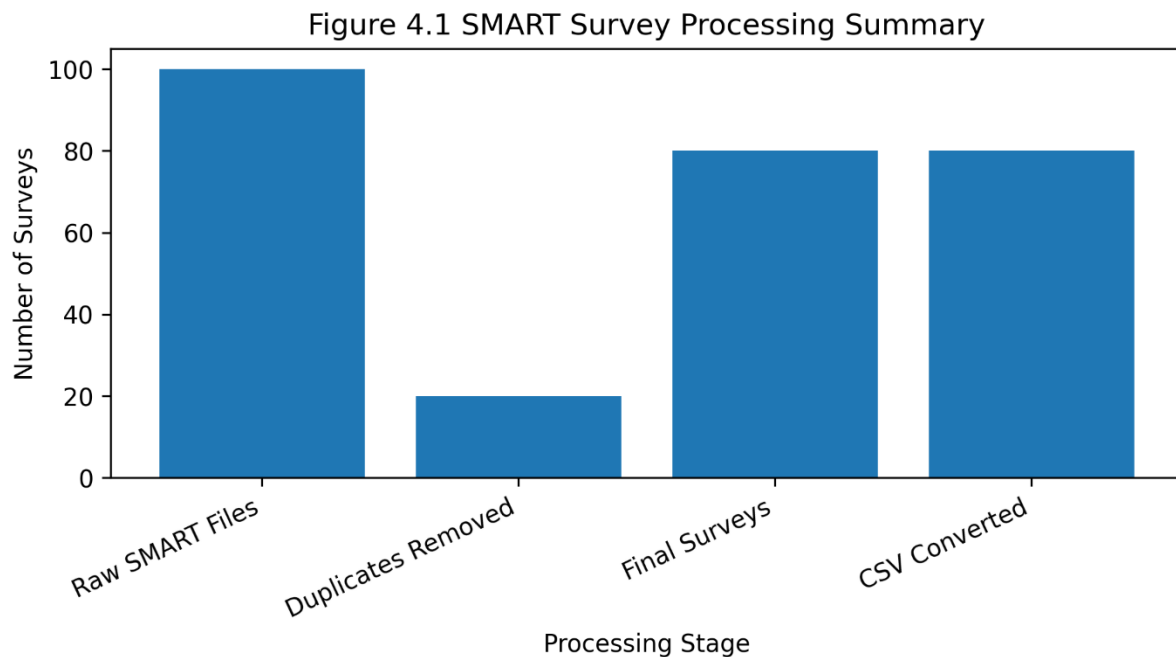
This chapter presents the results obtained from the SMART Survey Processing Pipeline and the machine learning models developed for mortality prediction. The results are organized according to the objectives of the study. First, the outcomes of the SMART survey processing stages are presented. Next, the characteristics of the final machine learning dataset are described. Finally, the performance of the three machine learning algorithms (K-Nearest Neighbors, Support Vector Machine, and Decision Tree) is presented and compared using Accuracy, Precision, Recall, F1-score, ROC-AUC, and Confusion Matrix results.

4.2 SMART Survey Processing Results

The first stage of the study focused on transforming raw SMART survey files into structured datasets suitable for machine learning analysis. A total of **100 SMART survey files** were inspected during the preprocessing stage. After reviewing the files, **20 duplicate survey files** were identified and removed. The remaining **80 unique surveys** were successfully processed and converted into CSV format.

Figure 4.1

SMART Survey Processing Summary



X-axis: Processing Stage

- Raw SMART Files
- Duplicate Files Removed
- Final Surveys
- CSV Converted

Y-axis: Number of Surveys

Interpretation

Figure 4.1 shows the overall performance of the SMART Survey Processing Pipeline. Initially, 100 SMART survey files were inspected. After removing 20 duplicate files, 80 valid surveys remained. All 80 surveys were successfully converted into CSV format without processing failures. This demonstrates that the developed pipeline was able to automatically process multiple SMART survey files and prepare them for further analysis.

Table 4.1

SMART Survey Processing Summary

Description	Value
Raw SMART Survey Files	100
Duplicate Files Removed	20
Final Surveys Processed	80
Successful CSV Conversion	80
Failed Conversions	0

Interpretation

Table 4.1 demonstrates that the processing pipeline successfully handled all valid survey files. Removing duplicate surveys reduced repeated observations and improved the reliability of the dataset. The absence of failed conversions indicates that the pipeline was effective in processing different SMART survey structures.

4.3 Final Machine Learning Dataset

After mortality extraction, cleaning, and integration with anthropometric information, the final dataset was prepared for machine learning analysis.

The final dataset contained:

- **43,874 observations**
- **16 predictor features**
- **1 target variable**

The target variable represented mortality status:

- 0 = No mortality
- 1 = Mortality

Table 4.2

Final Machine Learning Dataset Summary

Description	Value
Total Observations	43,874
Predictor Variables	16
Target Variable	1
Total Variables	17

Interpretation

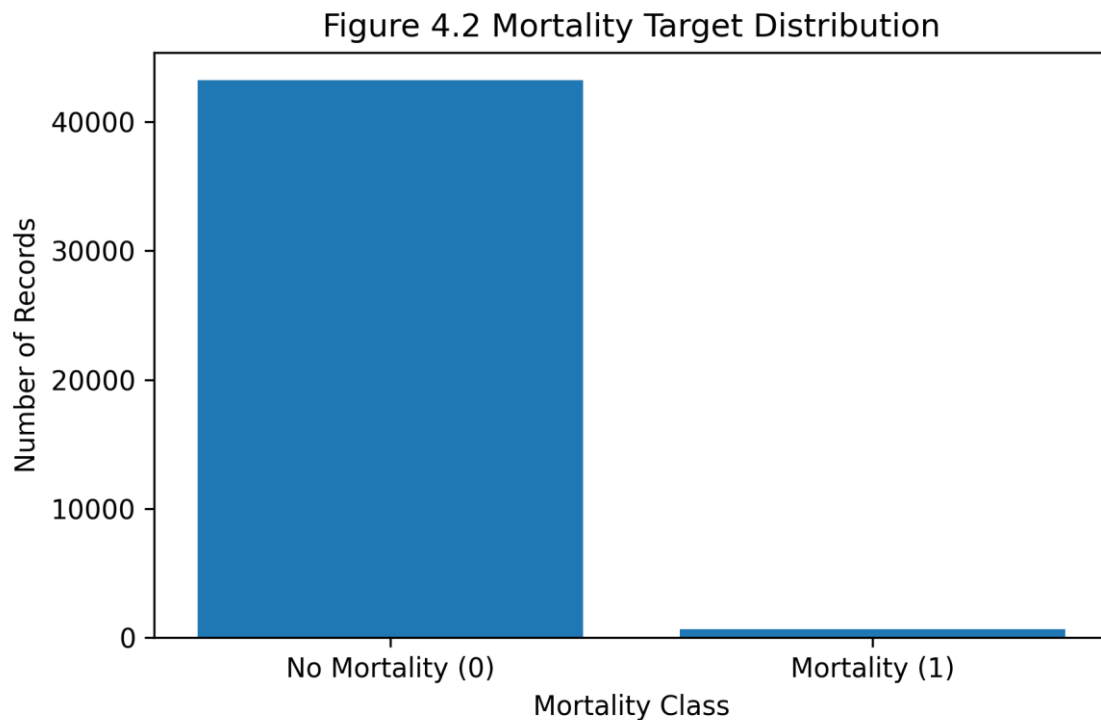
Table 4.2 shows that the preprocessing stage successfully produced a standardized dataset suitable for mortality prediction. The final dataset combined anthropometric and household-level information with the mortality target variable, allowing machine learning algorithms to learn patterns related to mortality outcomes.

4.4 Target Variable Distribution

The mortality prediction dataset contained a highly imbalanced target variable. Most observations belonged to the non-mortality class, while mortality cases represented a small proportion of the dataset.

Figure 4.2

Mortality Target Distribution



X-axis: Mortality Class

- No Mortality (0)
- Mortality (1)

Y-axis: Number of Records

Interpretation

Figure 4.2 shows the distribution of mortality and non-mortality observations. The dataset contained **43,219 non-mortality cases** and **655 mortality cases**. This imbalance creates a challenge for machine learning models because algorithms may focus on the majority class and fail to detect mortality cases. Therefore, SMOTE was applied to balance the training dataset before model development.

4.5 Machine Learning Model Performance Comparison

Three machine learning models were trained and evaluated:

- K-Nearest Neighbors (KNN)
- Support Vector Machine (SVM)
- Decision Tree

The models were evaluated using Accuracy, Precision, Recall, F1-score, and ROC-AUC.

Table 4.3

Machine Learning Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
KNN	0.8746	0.0189	0.1450	0.0335	0.520
SVM	0.7201	0.0166	0.3053	0.0315	0.535
Decision Tree	0.5501	0.0179	0.5420	0.0347	0.564

Interpretation

Table 4.3 shows that KNN achieved the highest accuracy; however, accuracy alone is not sufficient for evaluating mortality prediction because the dataset contains very few mortality cases.

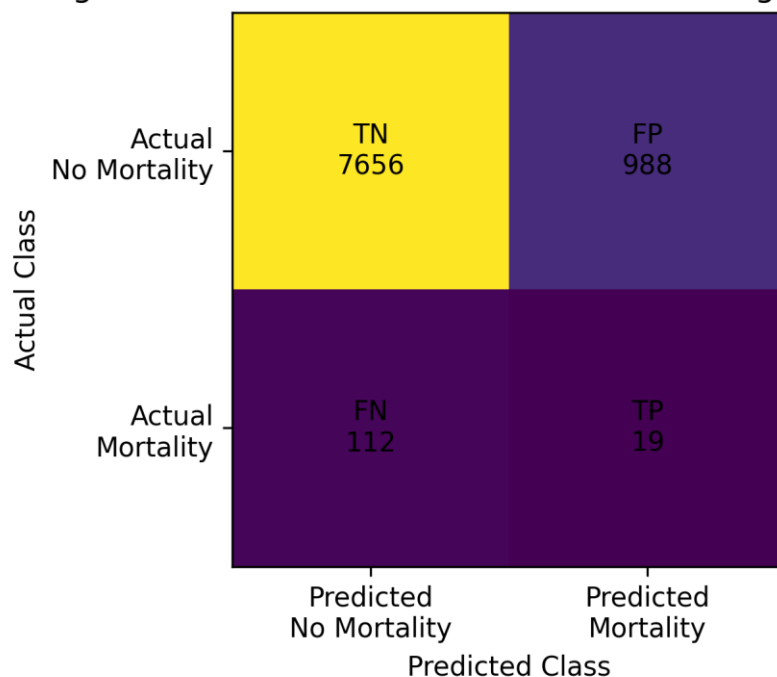
The Decision Tree model achieved the highest Recall (54.20%) and the highest ROC-AUC value (0.564). This indicates that Decision Tree was more effective at identifying mortality cases compared with KNN and SVM. Since detecting mortality cases is the main objective of this study, Recall was considered the most important performance measure.

4.6 Confusion Matrix Results

Figure 4.3

Confusion Matrix of K-Nearest Neighbors Model

Figure 4.3 Confusion Matrix of K-Nearest Neighbors Model

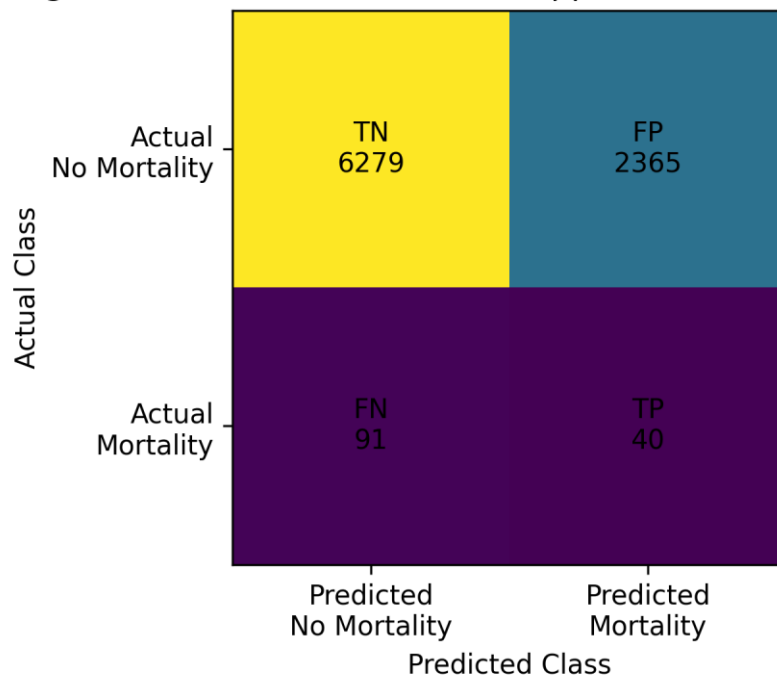


Interpretation

The KNN model correctly classified 7,656 non-mortality cases and identified 19 mortality cases. However, it failed to detect 112 actual mortality cases. Although the model achieved high accuracy, its ability to identify mortality cases was limited.

Figure 4.4
Confusion Matrix of Support Vector Machine Model

Figure 4.4 Confusion Matrix of Support Vector Machine Model

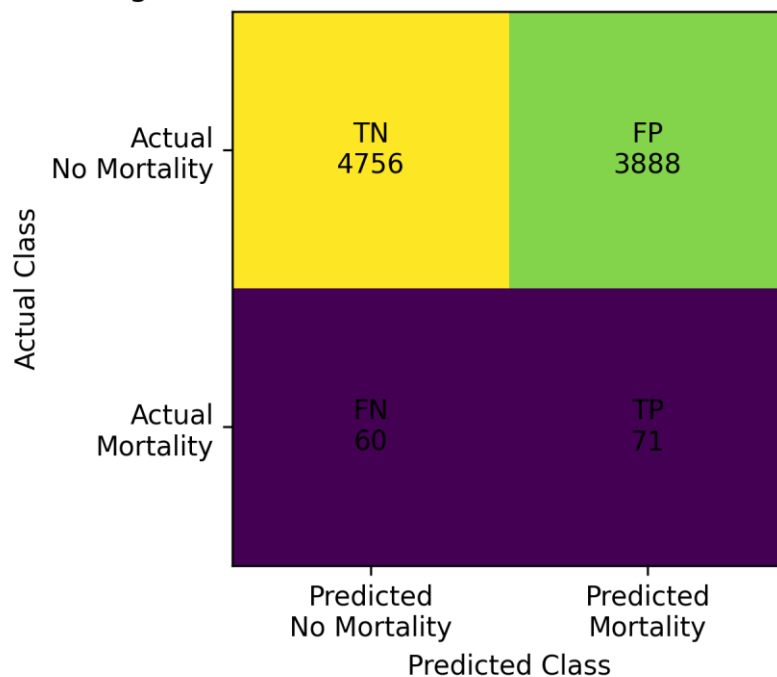


Interpretation

The SVM model correctly identified 40 mortality cases and missed 91 mortality cases. Compared with KNN, SVM achieved better mortality detection performance but produced more false positive predictions.

Figure 4.5
Confusion Matrix of Decision Tree Model

Figure 4.5 Confusion Matrix of Decision Tree Model



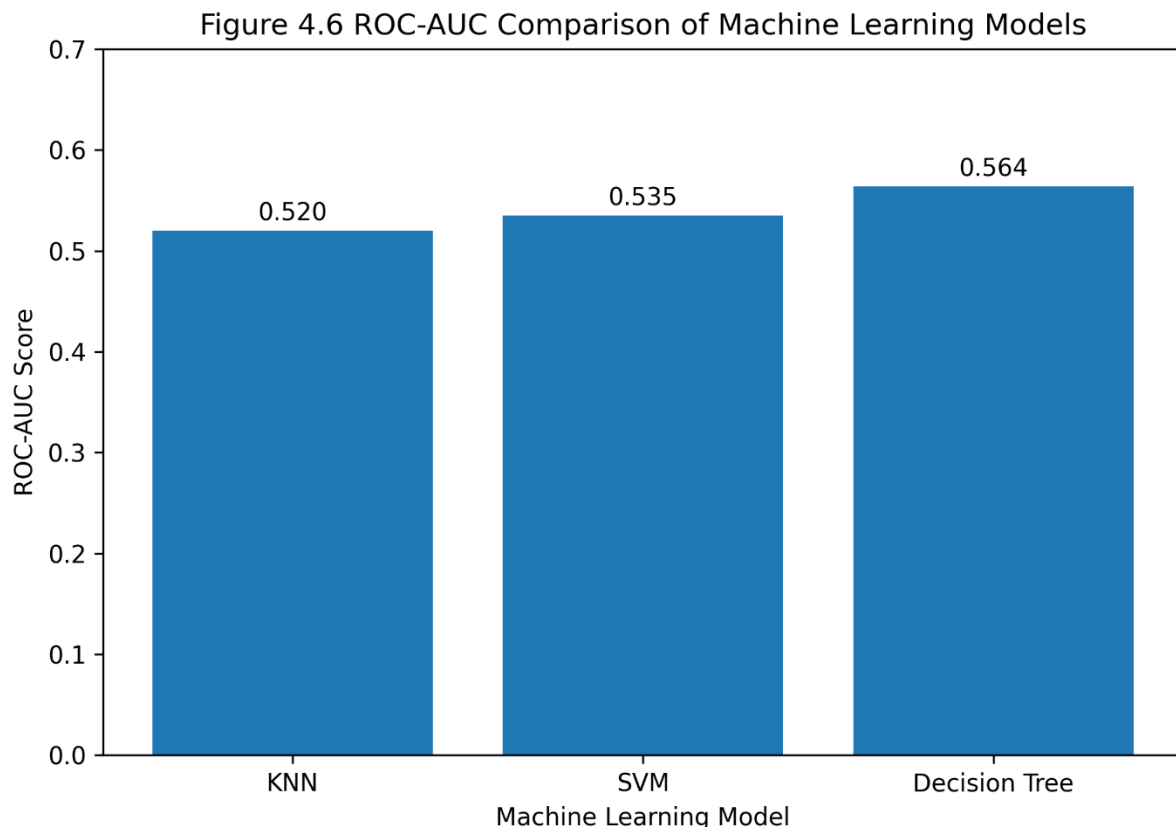
Interpretation

The Decision Tree model detected the highest number of mortality cases, correctly identifying 71 mortality observations. However, it also generated a higher number of false positive predictions. Despite this limitation, it provided the strongest mortality detection ability among the evaluated models.

4.7 ROC-AUC Comparison

Figure 4.6

ROC-AUC Comparison of Machine Learning Models



X-axis: Machine Learning Model

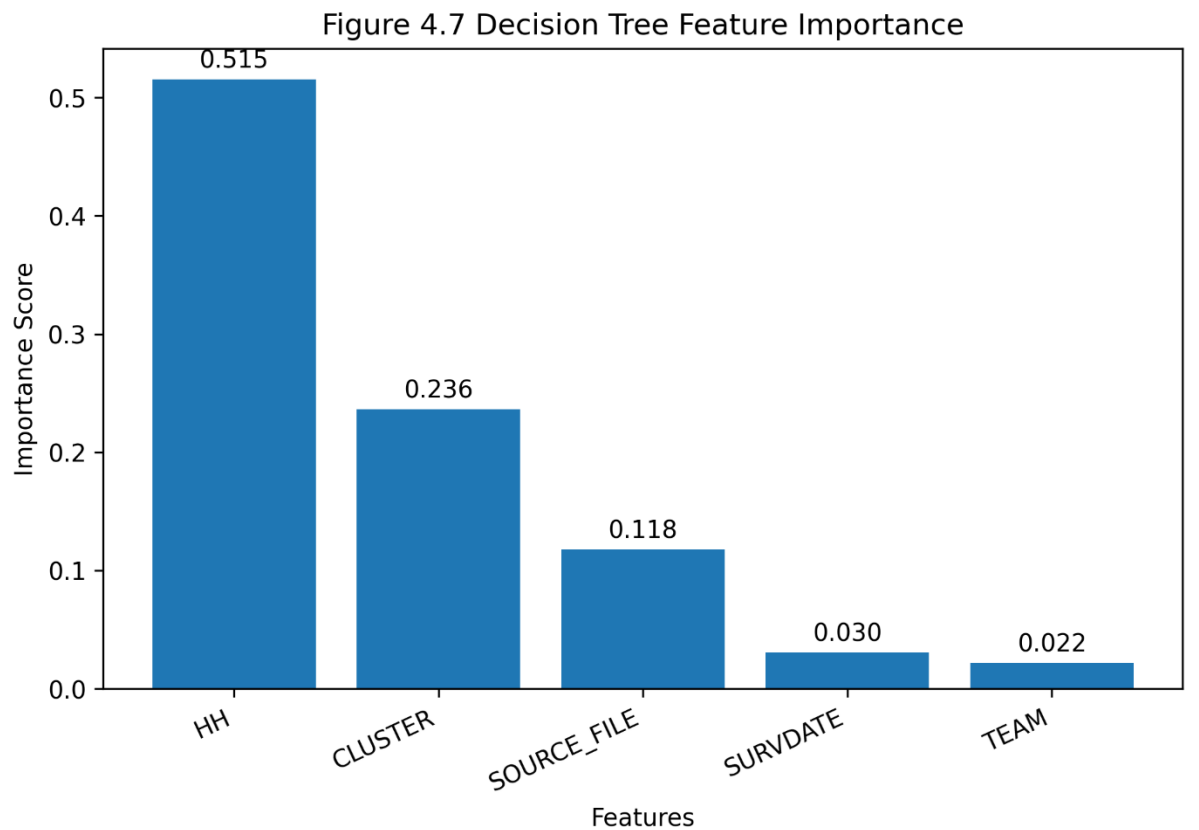
Y-axis: ROC-AUC Score

Figure 4.6 compares the ROC performance of the three machine learning models. The Decision Tree model achieved the highest ROC-AUC value (0.564), followed by SVM (0.535) and KNN (0.520). The results indicate that Decision Tree had the strongest ability to distinguish between mortality and non-mortality classes.

4.8 Feature Importance Analysis

Figure 4.7

Decision Tree Feature Importance



X-axis:

Features

Y-axis: Importance Score

Interpretation

Figure 4.7 shows the most important features used by the Decision Tree model. The highest contributing variables were:

- HH (0.515)
- CLUSTER (0.236)
- SOURCE_FILE (0.118)
- SURVDATE (0.030)
- TEAM (0.022)

These features contributed the most to mortality classification decisions. Anthropometric variables such as WAZ, HAZ, and WHZ also contributed, but with lower importance values.

4.9 Summary of Findings

The SMART Survey Processing Pipeline successfully transformed raw survey files into a machine learning-ready dataset. The pipeline processed 100 survey files, removed duplicate records, and produced a standardized dataset containing 43,874 observations.

The machine learning experiments showed that mortality prediction is challenging because mortality cases represent a small proportion of the dataset. Although KNN achieved the highest accuracy, Decision Tree provided the best mortality detection performance based on Recall and ROC-AUC.

Overall, the Decision Tree model was selected as the best-performing model in this study because it achieved the highest ability to identify mortality cases, which is the primary objective of mortality prediction.

5. Discussion

This study developed a SMART survey processing pipeline and evaluated machine learning models for mortality prediction using SMART survey data. The findings demonstrate that raw SMART survey files can be successfully transformed into a structured machine learning dataset through systematic preprocessing, cleaning, and integration steps.

The results showed that mortality prediction is a challenging task because of the highly imbalanced nature of the dataset. The number of mortality cases was considerably smaller compared with non-mortality cases. This imbalance affected model performance, particularly precision and F1-score. Although some models achieved high accuracy, this measure alone was not sufficient because a model could achieve high accuracy by mainly predicting the majority class.

Among the evaluated models, the Decision Tree model demonstrated the strongest performance for mortality detection. Although KNN achieved the highest accuracy (87.46%), it showed limited ability to identify mortality cases with a recall of 15%. In comparison, Decision Tree achieved the highest recall of 54.20% and the highest ROC-AUC value of 0.564. This indicates that Decision Tree was more effective at identifying households with mortality outcomes. In mortality prediction applications, identifying true mortality cases is more important than achieving high overall accuracy because missed cases may reduce the effectiveness of humanitarian interventions.

The feature importance analysis from the Decision Tree model showed that household-related and survey-location variables contributed significantly to prediction performance. The variables with the highest importance were HH, CLUSTER, and SOURCE_FILE. These findings suggest that differences between household structures and survey locations may influence mortality patterns. Anthropometric indicators such as WAZ, HAZ, and WHZ also contributed to the prediction process, indicating the importance of nutritional conditions in mortality risk assessment.

The findings of this study support the use of machine learning as an additional analytical approach for SMART survey data. Machine learning models can identify relationships between multiple variables that may not be easily detected through traditional analysis methods. Similar approaches have been applied in healthcare research to support risk prediction and decision-making using large health datasets (Rajkomar et al., 2019).

However, this study has several limitations. First, the mortality dataset was highly imbalanced because mortality cases represented a small proportion of total observations. Although SMOTE was applied to address this issue, synthetic samples may not fully represent real mortality patterns. Second, SMART surveys were collected from different locations and time periods, which may introduce variations in data quality and population characteristics. Third, the prediction performance remained moderate, suggesting that additional features and larger datasets may be required to improve future models.

Despite these limitations, the study successfully demonstrated that SMART survey data can be processed into a machine learning-ready format and used for mortality prediction. The developed pipeline provides a foundation for future improvements, including testing additional algorithms, incorporating more health-related variables, and applying advanced methods for improving prediction performance.

6. Conclusion

This study developed a complete SMART Survey Processing Pipeline and applied machine learning algorithms to predict mortality outcomes. The research successfully processed SMART survey files, extracted mortality and anthropometric information, cleaned and integrated datasets, and prepared a final machine learning dataset.

Three machine learning models were evaluated: K-Nearest Neighbors (KNN), Support Vector Machine (SVM), and Decision Tree. The models were compared using accuracy, precision, recall, F1-score, confusion matrix, and ROC-AUC metrics.

The results showed that mortality prediction from SMART survey data is challenging because of the limited number of mortality cases compared with non-mortality observations. Although KNN achieved the highest accuracy, Decision Tree provided the best mortality detection capability by achieving the highest recall and ROC-AUC performance.

The study concludes that machine learning methods can support mortality analysis from SMART survey data by identifying patterns associated with mortality outcomes. The developed pipeline improves the preparation and analysis of SMART survey datasets and provides a useful approach for future humanitarian health prediction systems.

Future work should focus on collecting larger datasets, improving feature selection, testing additional machine learning algorithms, and applying advanced techniques to increase mortality prediction accuracy and reliability.

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